

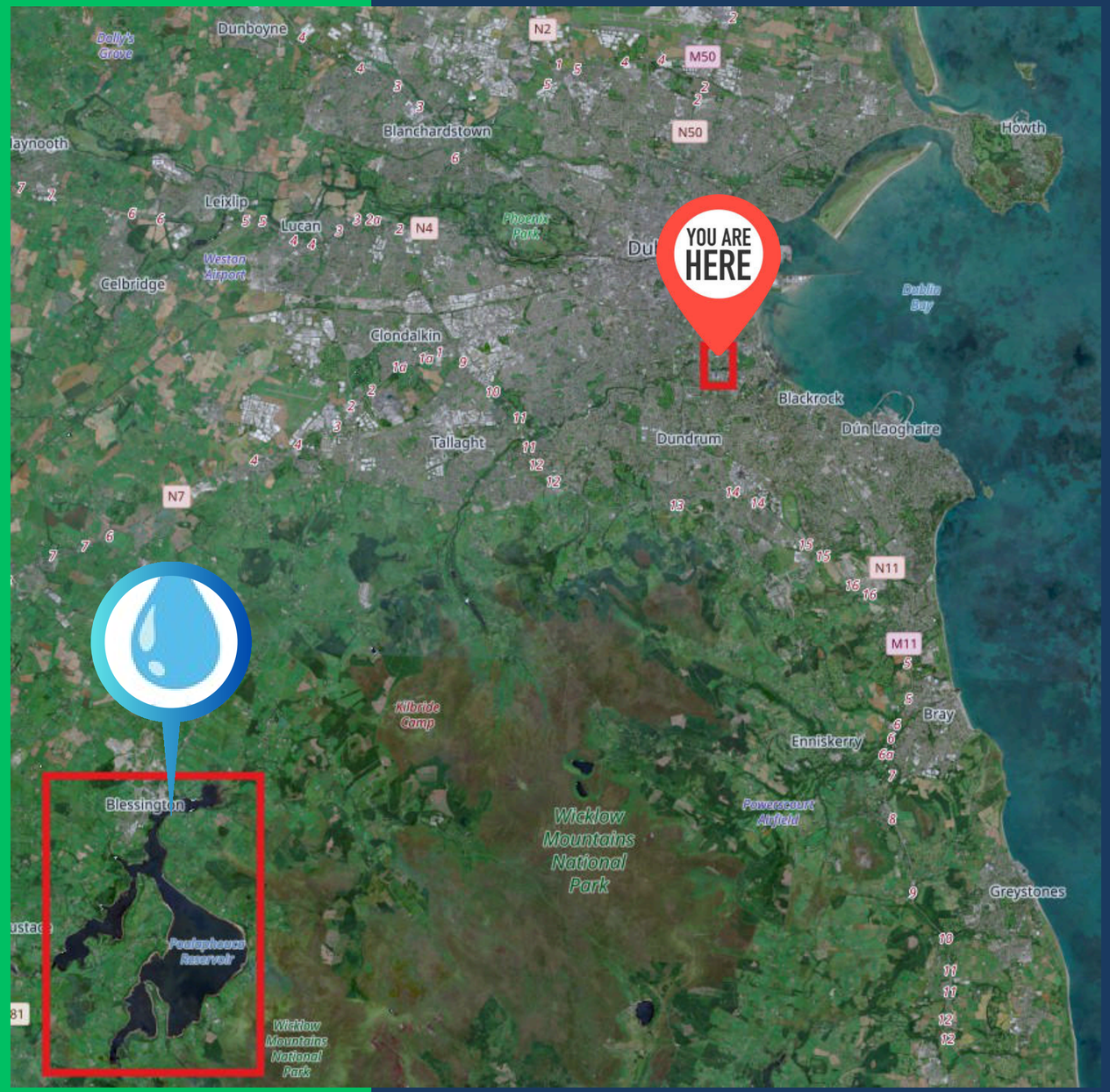
Hydrology InSAR Study

Blessington Lakes Elevation & Algae Analysis

Created & Presented by
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GEOL40830 – SAR and InSAR remote sensing

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Space Industry Skillnet

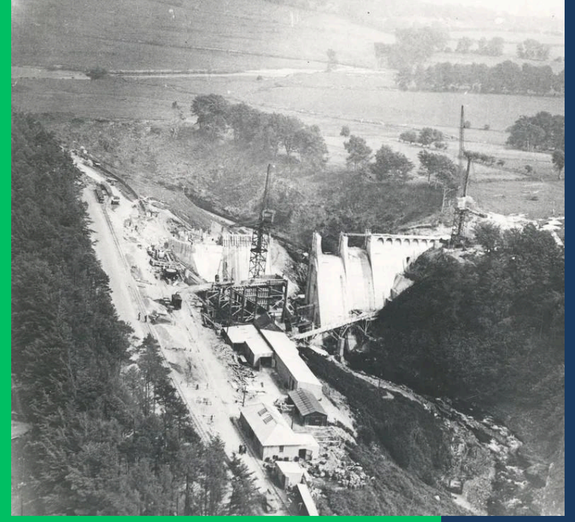


Introduction

Importance of Monitoring Irish Lakes

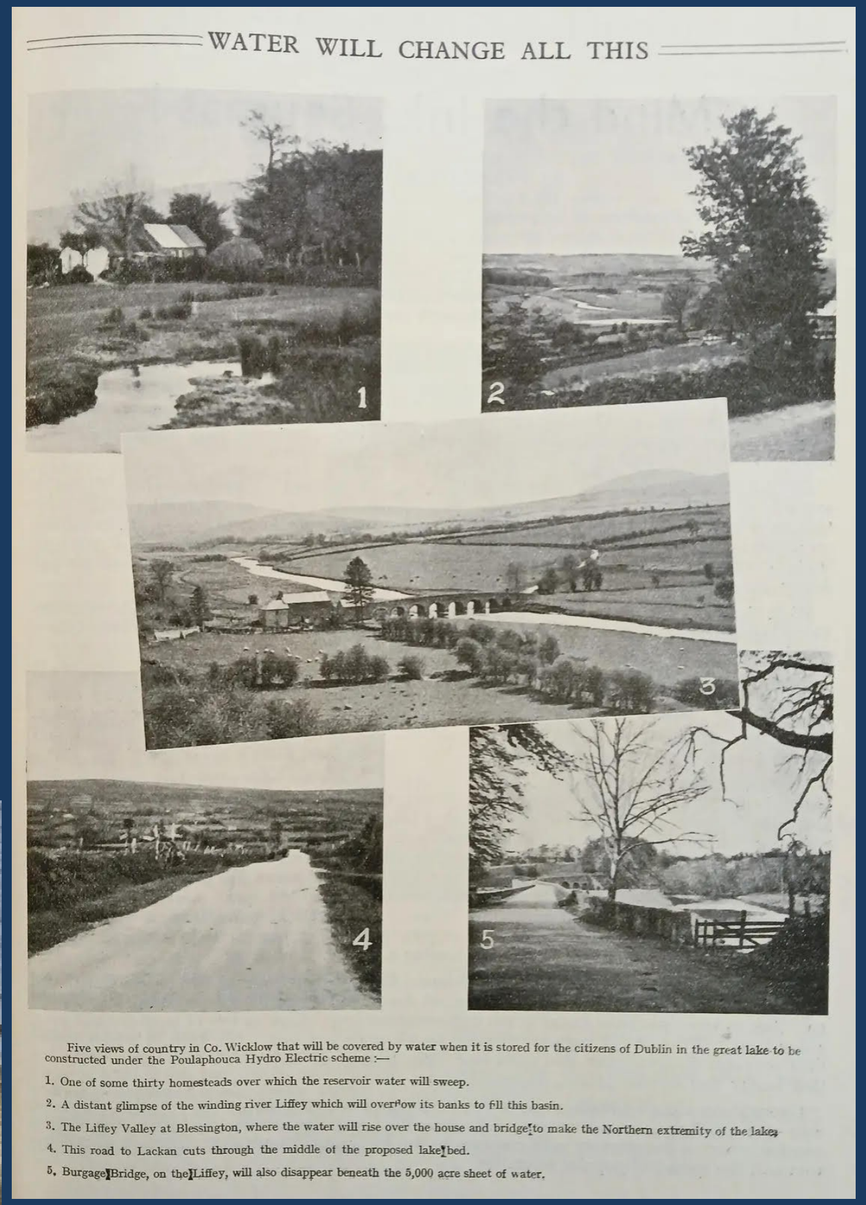
- Lake elevation and algae monitoring are essential for water management & health
- Irelands largest manmade water body; the Blessington Lakes, **provide just over 50% of Dublin's water supply**, making its availability and purity vital for both humans and animals alike.

But what if the lake experiences a dry season or drought?
What happens if algae blooms grow?
How do these factors effect the water quality & its availability?









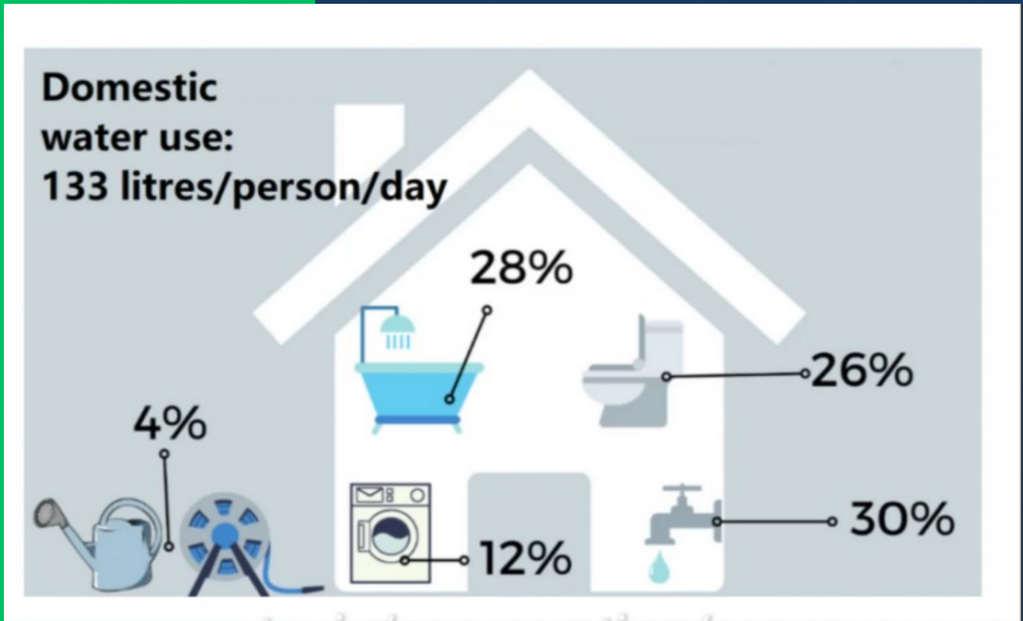
Five views of country in Co. Wicklow that will be covered by water when it is stored for the citizens of Dublin in the great lake to be constructed under the Poulaphouca Hydro Electric scheme :—

1. One of some thirty homesteads over which the reservoir water will sweep.
2. A distant glimpse of the winding river Liffey which will overflow its banks to fill this basin.
3. The Liffey Valley at Blessington, where the water will rise over the house and bridge to make the Northern extremity of the lake.
4. This road to Lackan cuts through the middle of the proposed lake bed.
5. Burgage Bridge, on the Liffey, will also disappear beneath the 5,000 acre sheet of water.

History of Lakes

'The Flooded Valley'

- Created in the 1930-40s by damming River Liffey to supply power and clean water to Dublin.
- Meant demolishing homes, farms and bridges - more than 70 families were told to relocate (Minimally compensated).
- Lake later became a huge attraction for Blessington area.
- Monitored with regular testing of nutrients & ecology,.
- Reports show wastewater and farming runoff contaminate lake.
- Long term protection and management essential!



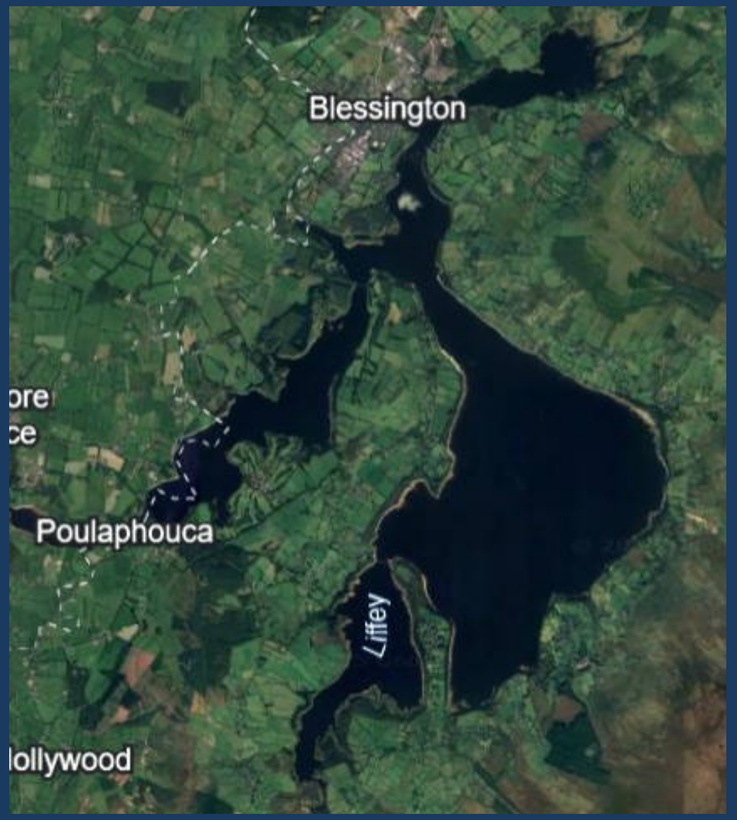


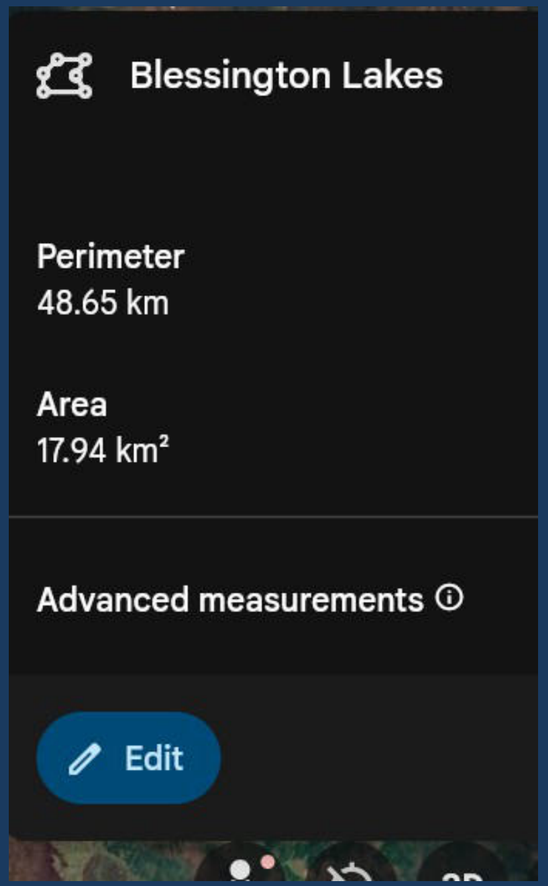
Dangers associated with ‘Doing Nothing’

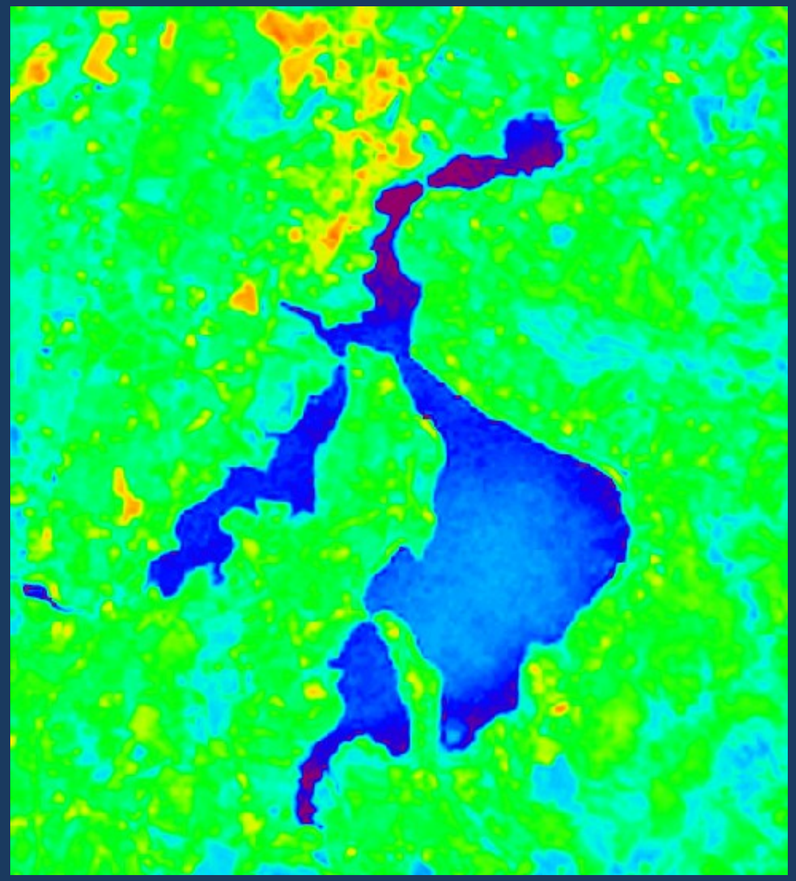
- The Blessington Lakes water reservoir is essential for Dublin's population, industry, and agriculture.
- Algae blooms degrade water quality, raise treatment costs and threaten human, marine and bio health.
- Drought worsens these problems by reducing water levels and fringing the waters availability.

Data Acquisition Understanding InSAR Data Processing

The process involved obtaining remote sensing datasets, followed by data calibration, noise reduction, and georeferencing to ensure accurate elevation measurements and algae detection for Blessington Lake's water. The data was acquired using SWOT and SENTINEL data.









Elevation Extraction

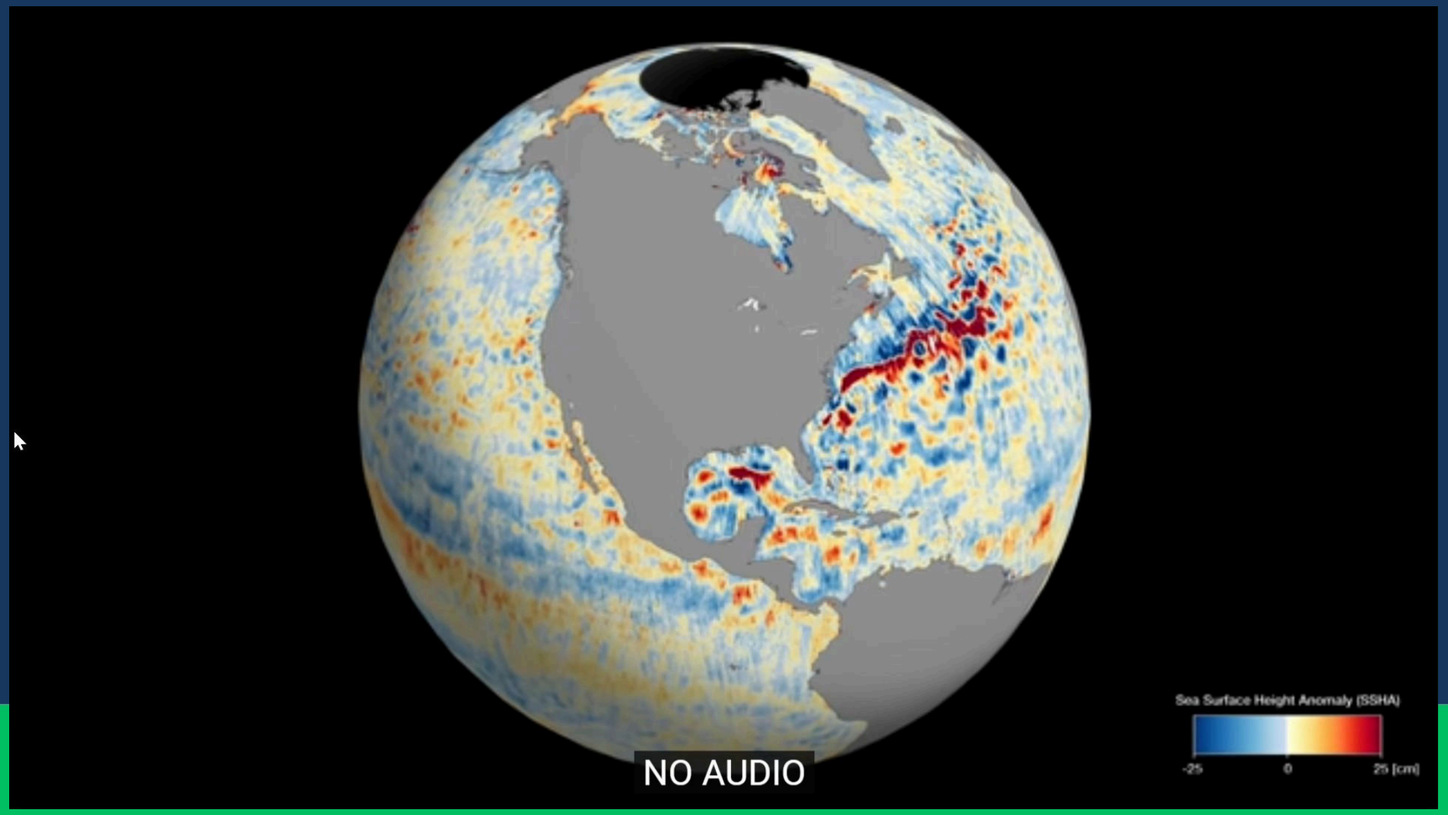
InSAR SWOT techniques

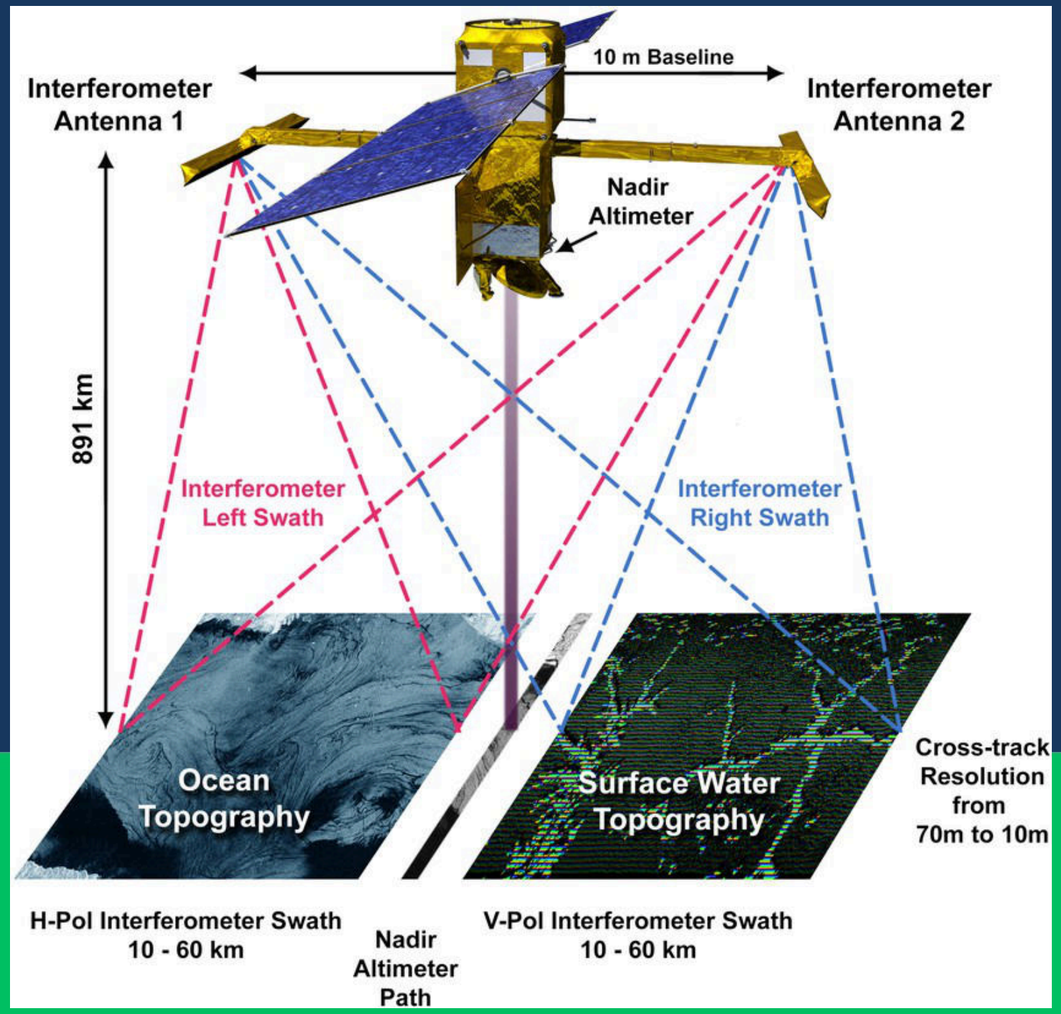
An interferometer bounces radar pulses off the surface of water and receives return signals with two different antennas at the same time, this allows SWOT to triangulate the height of the water's surface. This provides accurate lake elevation data and can be used for detailed monitoring of water level changes over time through wet and dry seasons with high precision.



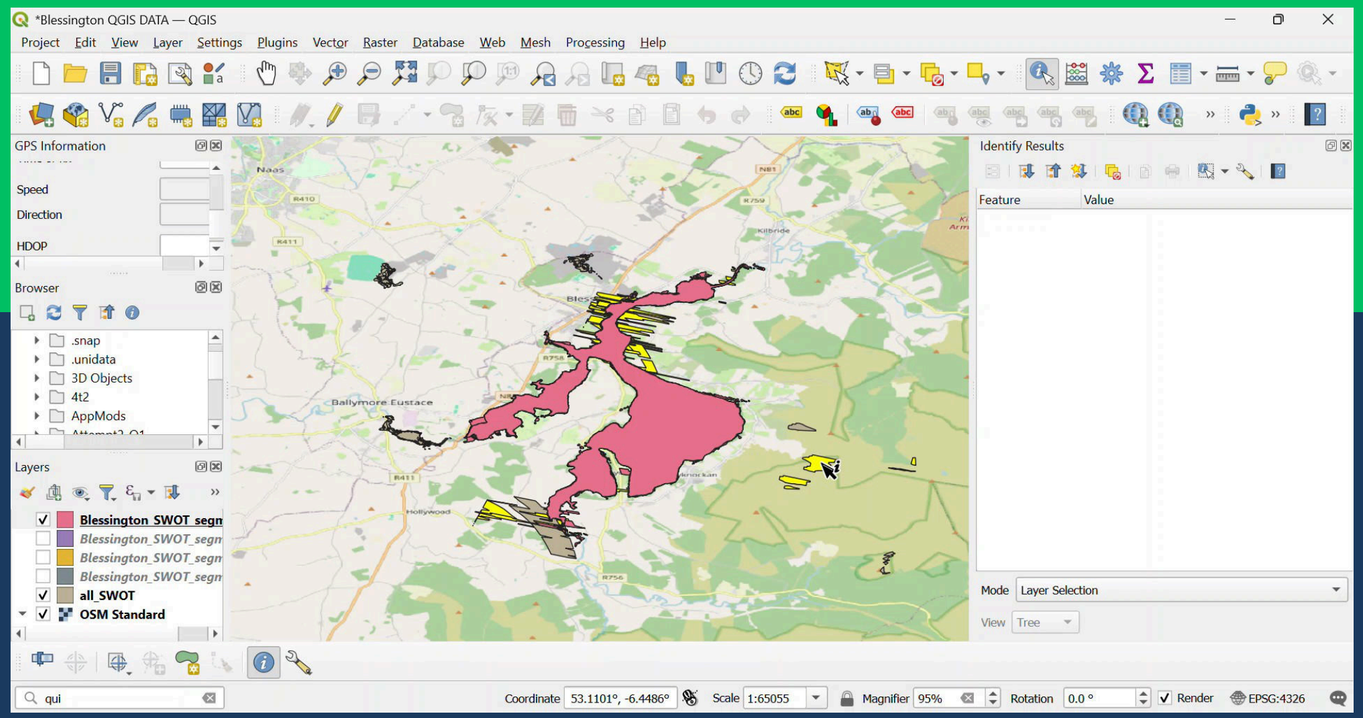
NOTES ON ANALYSIS

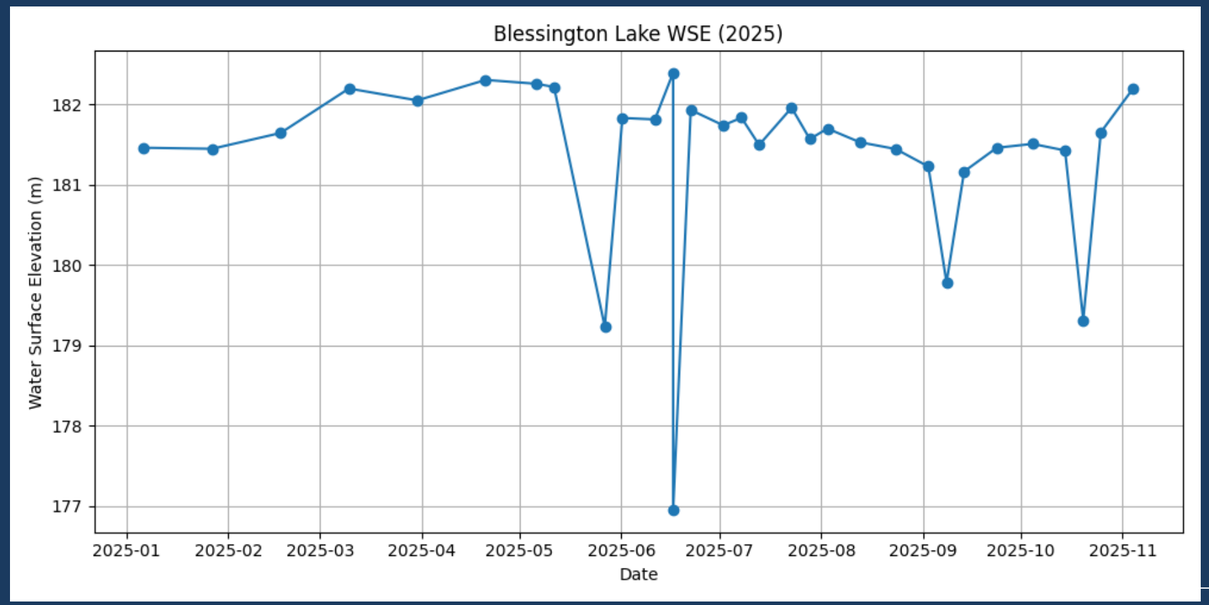
Elevation Extraction





Results; Water Level Changes





Video visualises the **fluctuating water levels** of Blessington Lake, highlighting significant changes due to seasonal precipitation patterns. This time series data was extracted to plot lake water elevation levels with time



Water Level Changes

Causes of Water Level Variability

The observed water level changes can be attributed to various factors;

- **Seasonal rainfall patterns,**
- **Rare drought conditions**,
- **Human interventions (nearby reservoir)**.

Highlighting the necessity for ongoing monitoring and analysis.

Algal Presence

Analyzing Water Levels Across Blessington Lake

- **What is it?**

Rapid growth of chlorophyll organism groups in water.

- **Causes:**

Warm temperatures and nutrient run-off.

- **Impact:**

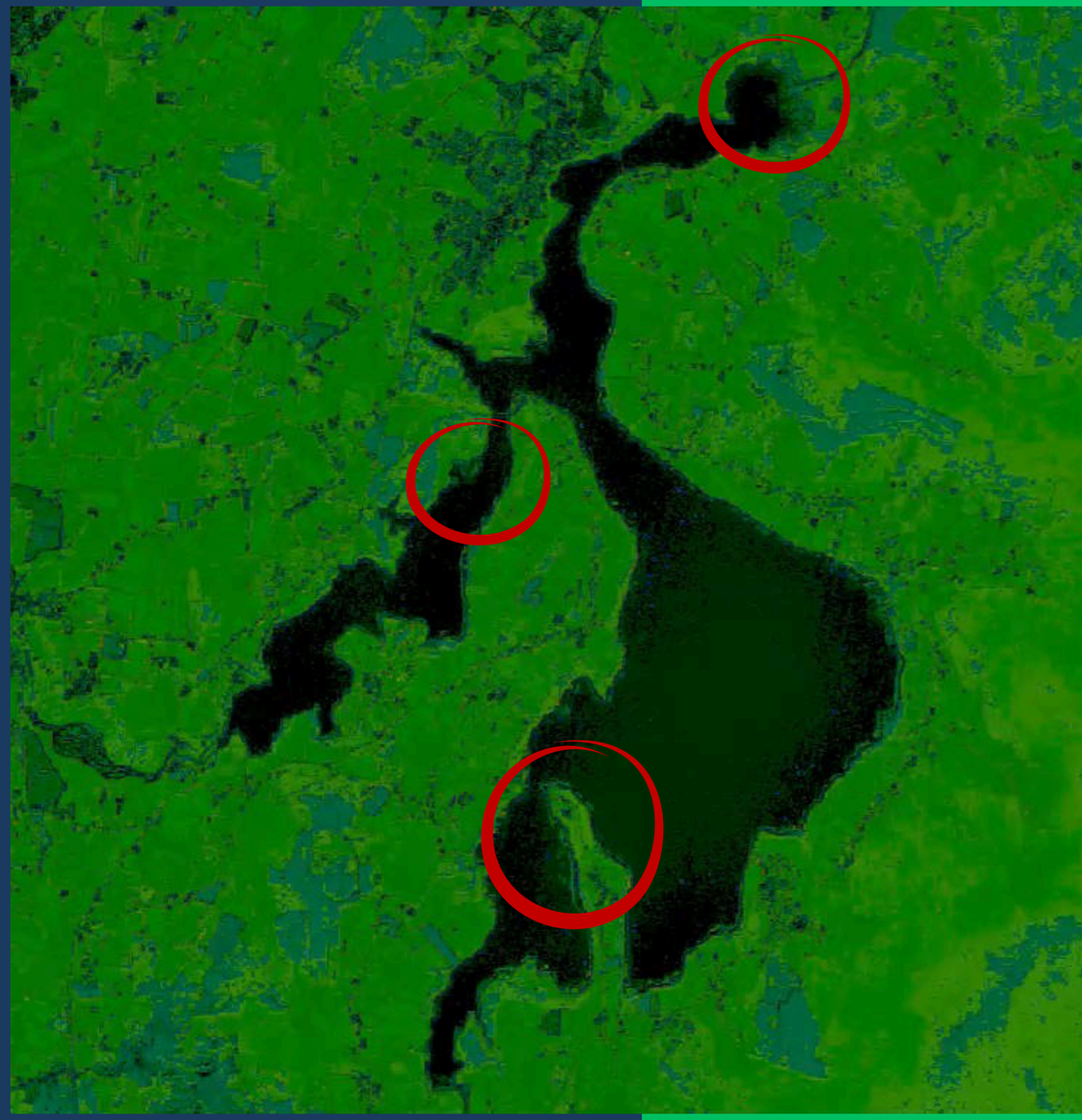
Can reduce water quality and affect wildlife.

- **Current Situation in Blessington:**

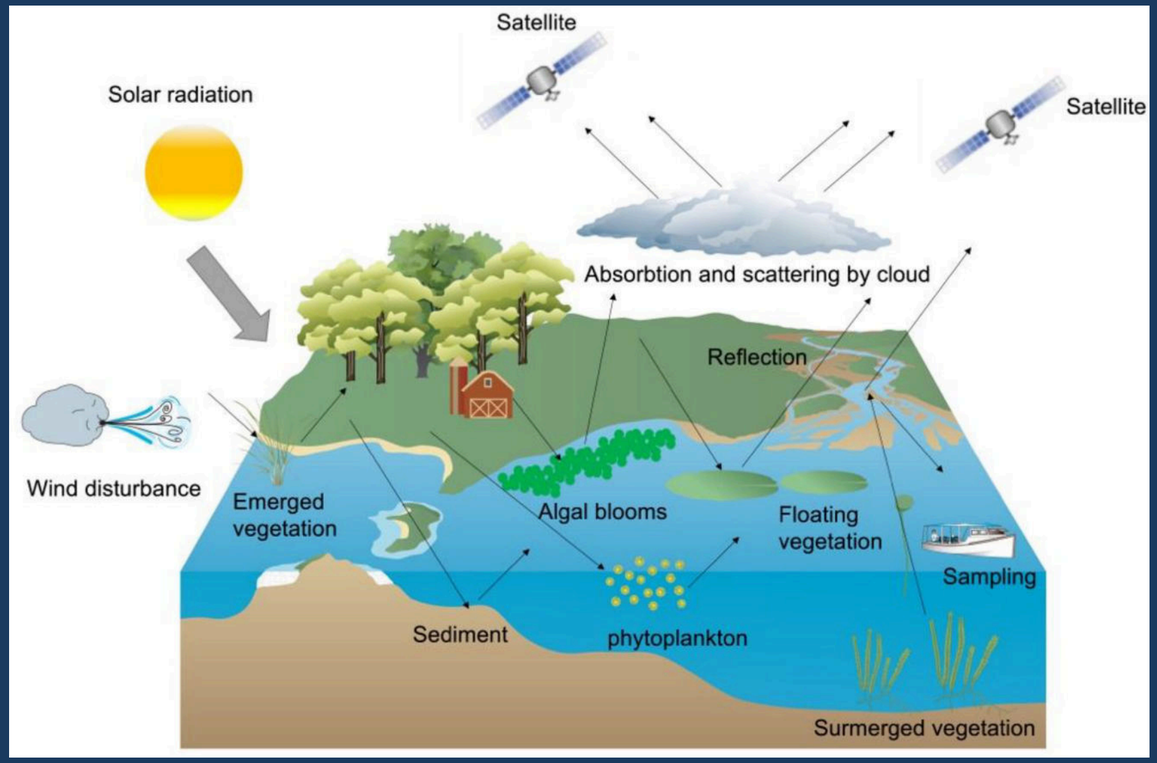
Mild bloom observed; monitoring ongoing.

- **Key Message:**

Stay cautious with water activities until conditions improve.



Algae Detection

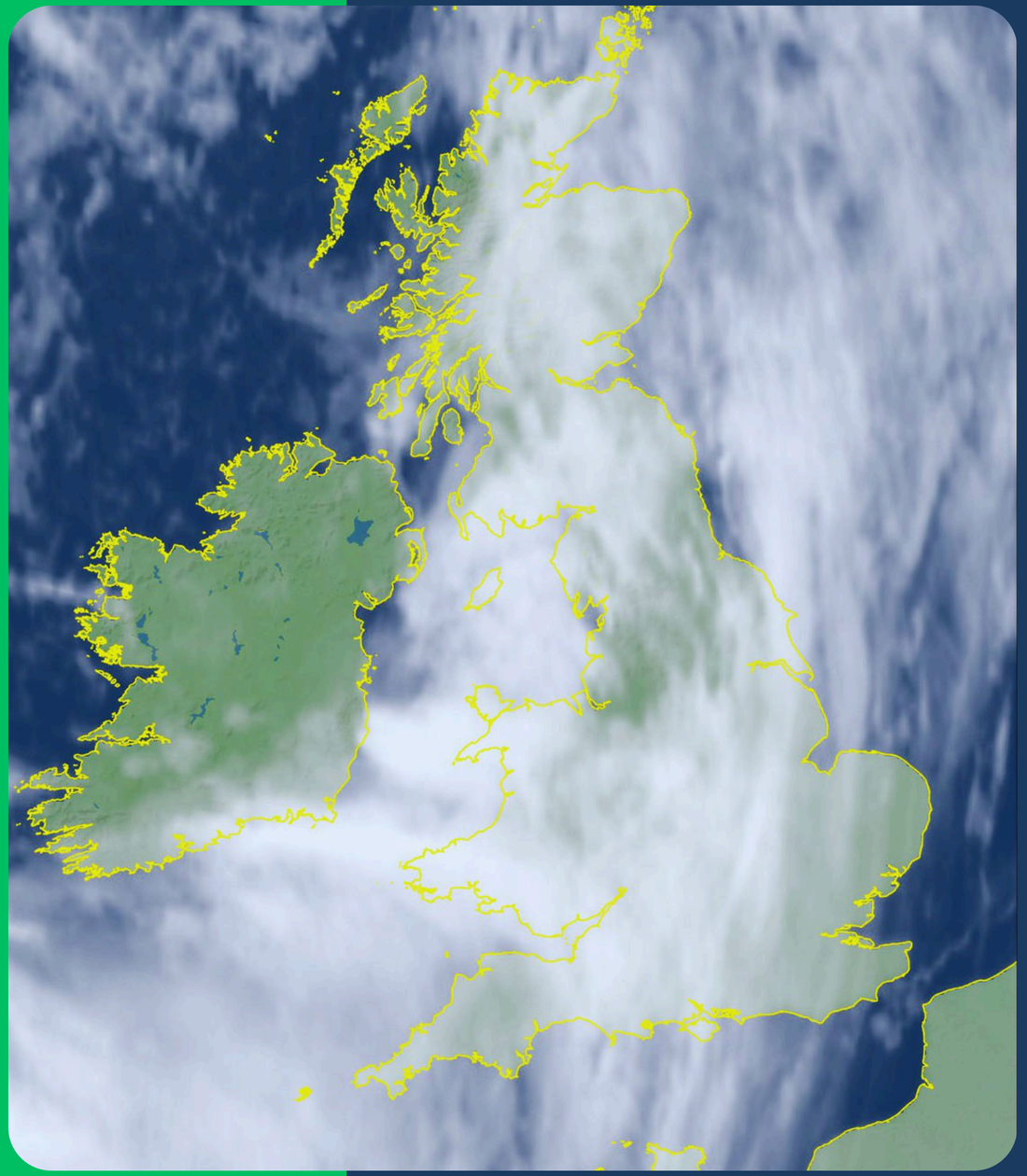








- **Method:** Optical
- Using surface-reflectance **Sentinel-2** images
- Calculated an algae index (B4 with B5/B6)
- Classified index values to map bloom intensity
- Compared early and later dates to track bloom development



Limitations and Future Work

Challenges and potential improvements

- The limitations of SAR/InSAR and optical remote sensing methods include **temporal resolution** and **atmospheric disturbances**.
- Future research should focus on enhancing accuracy and exploring additional data sources (such as SWOT) for comprehensive lake monitoring.

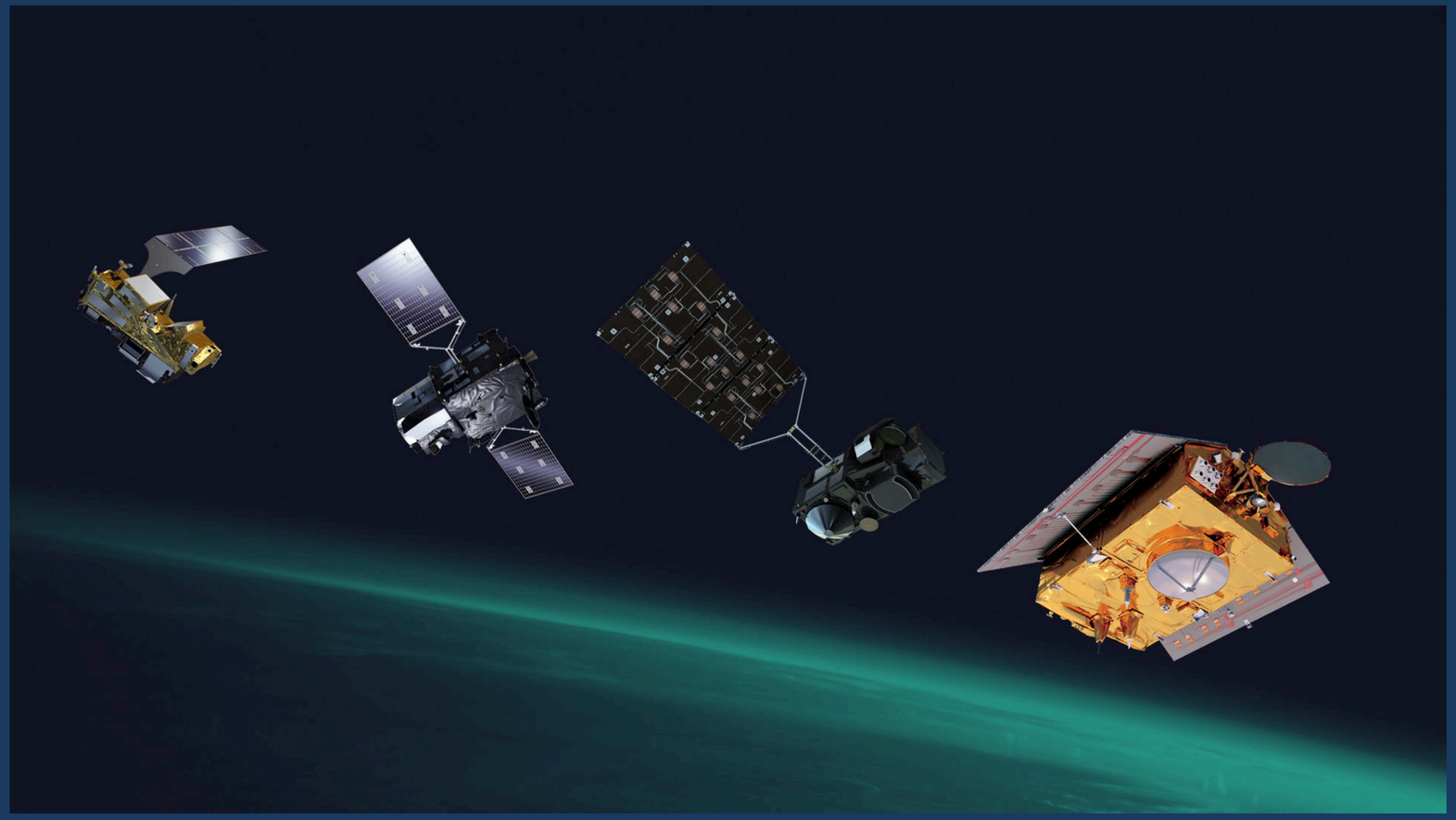


Conclusion

Successful analysis using Optical and InSAR

This study successfully demonstrated the utility of InSAR technologies for monitoring lake elevation and algal changes within the Blessington Lakes, emphasising the importance and uses of remote sensing in environmental management.





Thank you for listening.
Any questions?

Image References

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